

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461231.7459 +/- 0.009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.04
Transit depth (Rp/Rs)^2	1.69 +/- 1.03 [%]
Orbital Inclination (inc)	84.67 +/- 2.85
Airmass coefficient 1 (a1)	1.74 +/- 0.71
Airmass coefficient 2 (a2)	-0.26 +/- 0.11
Scatter in the residuals of the lightcurve fit is	36.88 %
Best Comparison Star	None
Optimal Aperture	2.73
Optimal Annulus	14.77
Transit Duration (day)	0.052 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.13 $\pm$ 0.04 vs 0.119 (deviation 0.25 σ)
Duration fit vs published	0.052 d vs 0.07 d (deviation 0.47 σ)
Mid-time O-C	7.8 min
Depth SNR	2.9
Residual scatter	36.88 %

Light curve

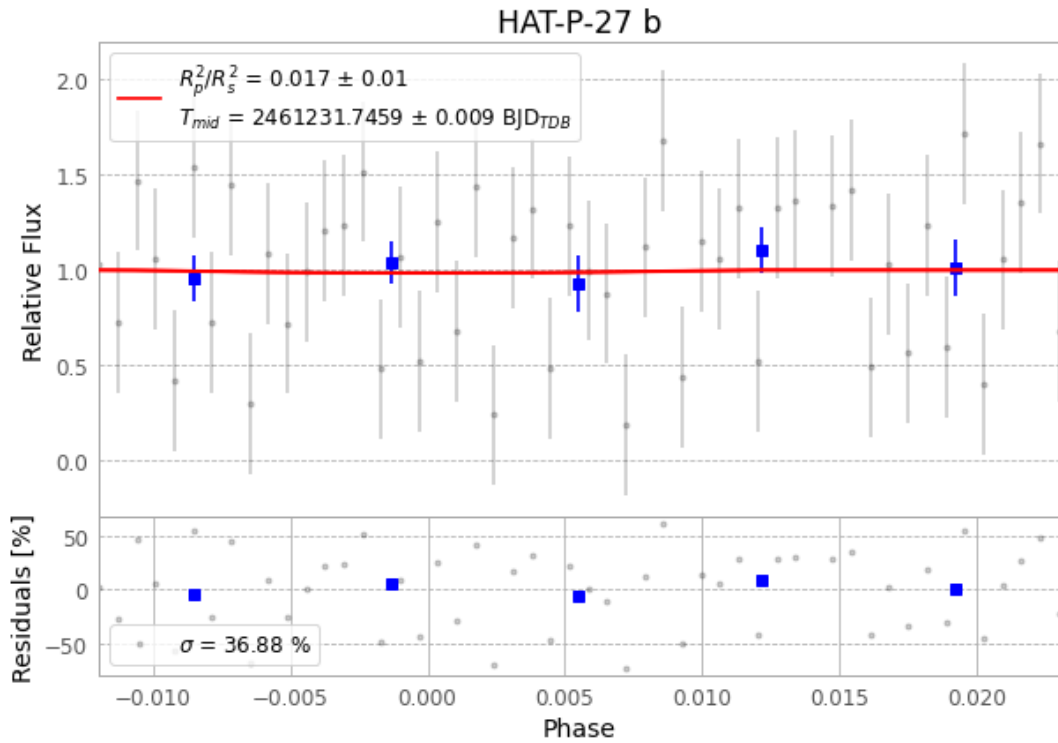


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-07-09.png (SHA-256 616aff2ab666aa87...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dbed0536df314674...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), HATP-27260710045715.FITS → HATP-27260710073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260710.log (c42aea1b6117...), FinalLightCurve_HAT-P-27 b_2026-07-09.png (616aff2ab666...), FinalLightCurve_HAT-P-27 b_2026-07-09.csv (1777868f142c...)

Compute resources

Wall time 146.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:32Z.